



Security Concerns and Risk related to Cryptocurrencies

Dr. Kaleem Usmani



Bitcoin is the beginning of something *great*: a **currency** without a **government**, something *necessary* and ***imperative***.

- Nassim Taleb

A Cryptocurrency is a digital currency that is created through mathematical engineering (algorithm). It is designed to be open, anonymous, secure, fast and bypasses traditional financial structures.



Bitcoin, created in 2009, was the world's first Cryptocurrency. Since then, many new Cryptocurrencies (also known as Altcoins) have been introduced



ADVANTAGES OF CRYPTOCURRENCY OVER TRADITIONAL MONEY

Currency

VS

Traditional

Digital currency maintains its users complete anonymity. When you make a purchase with traditional money your personal information is attached to each and every transaction which can be used to track you and take note of your purchases. But cryptocurrency transactions carry no personal information.



Cryptocurrencies aren't directly linked to the laws, rules or regulations of any government, corporation or bank. Hence, the interest rates, fees and surcharges that you may have to pay on your bank account or credit card do not effect your transactions or cryptocurrency in any manner.



Accounts that hold traditional currency can be garnished or frozen completely. Cryptocurrencies, on the other hand, are not stored in traditional banks.



How does cryptocurrencies work?

The Mining Process



SERVERS GENERATE COMPLEX
MATH PROBLEMS

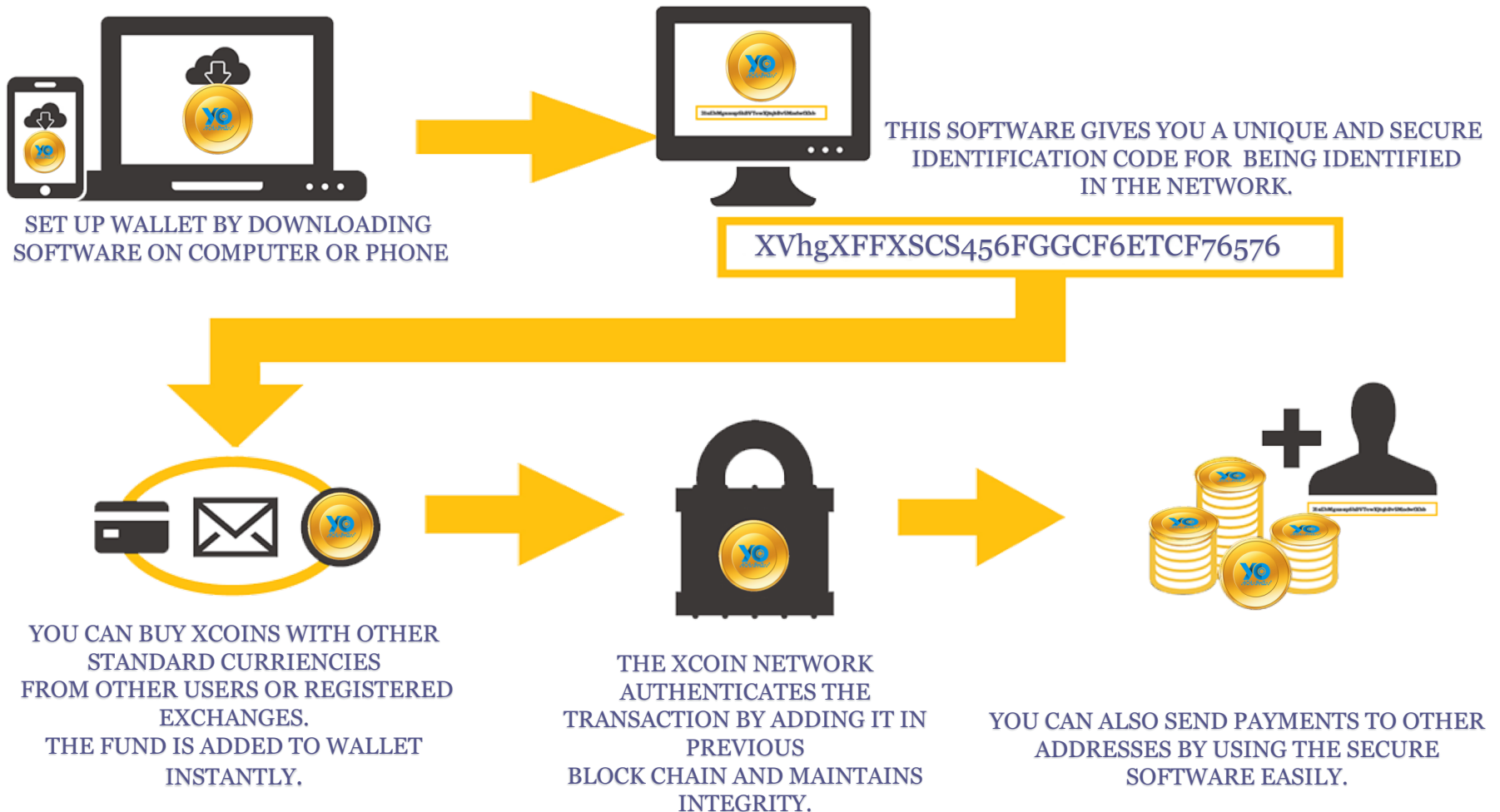


MINERS GENERATE SOLUTION



TO RELEASE XCOINS

How to use Cryptocurrencies



Role of Blockchain in Xcoin mining

USER 'A' SENDS FUND TO USER 'B'



THE TRANSACTION IS CONVERTED TO A 'BLOCK'



THE TRANSACTION IS BROADCASTED ON THE NETWORK FOR VALIDATION



THIS BLOCK IS ADDED TO THE EARLIER NON REVERSIBLE PROCESS CHAIN OF BLOCK

FINALLY USER 'B' RECIEVES FUNDS FROM USER 'A'



Security Concerns with Cryptocurrencies

Selfish Mining— This allows a sufficient size pool of “selfish miners” to gain revenue larger than its ratio of mining power, which forces “honest miners” to spend their cycles on blocks that won’t make it to the blockchain.

Double Spending— This allows an attacker to successfully make more than one transaction using a single coin, which invalidates the “honest” transaction.

Security Concerns with Cryptocurrencies

Wallet Software/Distributed Denials of Service Attacks (DDoS)—“Wallets” are client-side applications used to manage Bitcoins and transactions of Bitcoins from/to the client and can be accessed online or via download. Online wallets are more vulnerable to DDoS attacks since they need encryption and are backed off-line.

Acquiring Greater Than 50% Computing Power— This is when any conspiring user acquires more than 50% of the computing power in mining process, which can also lead to other attacks.

Security Concerns with Cryptocurrencies

Timejacking— This happens when an attacker announces an inaccurate timestamp while connecting to a node for a transaction, altering the network time counter and deceiving the node, which can cause double-spending.



Thank You